

Manifold-Constrained Hyper-Connections for Stable Depth Scaling in LSTM Time-Series Forecasting

Chandan Kumar Shah, Abhaya Shrestha, Asmit Khanal, Kushal Regmi, Nabina Thapa
Department of Electronics and Computer Engineering

Pulchowk Campus, Institute of Engineering, Tribhuvan University, Lalitpur, Nepal

Email: 080bct023.chandan@pcampus.edu.np, 080bct006.abhaya@pcampus.edu.np, 080bct017.asmit@pcampus.edu.np,
080bct042.kushal@pcampus.edu.np, 080bct047.nabina@pcampus.edu.np

Abstract—Deep stacked LSTMs hit a depth ceiling: gradients vanish or explode and deep variants collapse to near-constant predictions. Hyper-Connections (HC) widen the residual stream with learnable mixing matrices but are unstable; manifold-constrained HC (mHC) restores stability for large language models via a doubly-stochastic (Birkhoff-polytope) projection. We adapt mHC to recurrent forecasting as a *macro* inter-layer highway that keeps the Sinkhorn projection outside the temporal loop, preserving cuDNN-optimized recurrence. On NOAA JFK weather and ETTh1, across depths $L \in \{4, 8, 16\}$ with five seeds, mHC keeps held-out test error stable as depth increases, while Standard LSTM/TCN collapse at depth and unconstrained HC diverges. Additional $L=32$ runs on weather and ETTh1 support the same depth-stability trend but have smaller seed counts. mHC keeps the composite residual spectral norm within numerical tolerance of 1.0, whereas HC’s exceeds 10^5 . We analyze the macro design’s complexity advantage over a per-timestep (micro) placement: the Sinkhorn projection runs $O(L)$ times rather than $O(T \cdot L)$. In short, the manifold constraint removes the LSTM depth ceiling.

Index Terms—time-series forecasting, LSTM, hyper-connections, doubly stochastic matrices, depth scaling, gradient stability.

I. INTRODUCTION

Recurrent networks, and LSTMs [1] in particular, remain competitive for multivariate time-series forecasting, yet *deep* stacked LSTMs are difficult to train: as layers accumulate, gradients vanish or explode [2] and the network often collapses to a near-constant predictor. In our experiments a 16-layer stacked LSTM on weather data degrades from a test MSE of 92.6 at $L=4$ to 204.4 at $L=16$, and a temporal convolutional network (TCN) collapses similarly. This “depth ceiling” limits the representational capacity that practitioners can actually use.

Residual connections [3], [4] mitigate this through an identity mapping that preserves signal across layers. Hyper-Connections (HC) [5] generalize residuals by widening the stream into n parallel channels with learnable mixing matrices, improving expressivity but breaking the identity mapping and, as we confirm, diverging at depth. Manifold-constrained HC (mHC) [6] restores stability for

large language models by projecting the residual-mixing matrix onto the Birkhoff polytope of doubly-stochastic matrices via the Sinkhorn–Knopp algorithm [7], which guarantees a non-expansive, compositionally closed mapping.

We adapt mHC to recurrent forecasting. Rather than applying the projection inside the recurrent cell (the per-token placement used for transformers), we introduce a *macro* inter-layer highway: the doubly-stochastic mixing operates between LSTM layers, outside the temporal loop, so the cuDNN-optimized sequence kernel is preserved and the Sinkhorn projection runs $O(L)$ times rather than $O(T \cdot L)$.

Contributions.

- A macro-level mHC highway for stacked LSTMs that decouples cross-stream mixing (doubly-stochastic) from temporal recurrence (cuDNN LSTM).
- Evidence on two five-seed datasets (NOAA JFK weather and ETTh1) across $L \in \{4, 8, 16\}$, plus measured $L=32$ extensions on weather and ETTh1, that mHC preserves held-out test error and bounded gradients/spectral norm at depth while standard LSTM/TCN collapse and unconstrained HC diverges.
- A controlled complexity analysis showing the macro placement reduces the number of Sinkhorn projections from $O(T \cdot L)$ (per-timestep/micro) to $O(L)$ (per-layer), making the manifold constraint practical for long sequences.

II. RELATED WORK

Residual and identity mappings. Deep residual networks [3] and the identity-mapping analysis [4] established that a clean additive skip path is critical for trainable depth. Our work asks how to retain that property while allowing richer cross-stream mixing.

Hyper-Connections and mHC. Hyper-Connections [5] widen the residual stream and learn how to route information across channels, but the unconstrained mixing matrices destroy the identity mapping and destabilize deep training. mHC [6] constrains the mixing matrix to the Birkhoff polytope

via Sinkhorn–Knopp [7], demonstrated at LLM scale on the DeepSeek-V3 architecture [8]. The mHC paper itself frames the choice of where to apply the highway as a *macro* (inter-block) vs. *micro* (intra-block) design decision [6]. We transfer this idea to recurrent forecasting and study *where* the projection should live for an LSTM: as a macro inter-layer highway, leaving the cuDNN-optimized sequence kernel intact. Concurrently, mHC has also been adapted to state-space language models [9]; that work extends mHC to SSMs for language modeling, whereas we focus on LSTM-based time-series forecasting and provide a controlled depth-scaling study.

Time-series forecasting baselines. Linear models such as DLinear/NLinear [10] are strong, simple baselines; recurrent (GRU [11]) and convolutional (TCN [12]) models are standard neural baselines; and transformer forecasters including Informer [13], PatchTST [14], iTransformer [15], and TimesNet [16] represent recent state of the art. We compare against classical, linear, recurrent, convolutional, and (PatchTST, iTransformer) transformer baselines under one protocol; extending to further transformer SOTA is future work (Sec. VI).

State-space models. Selective state-space models such as Mamba [17] offer an orthogonal route to stable, efficient long-sequence modeling via input-dependent recurrences rather than a wider residual stream. We do not train an SSM baseline in this work; mHC is a modification to the *inter-layer highway* around an existing recurrent cell (here, the LSTM) and is in principle compatible with an SSM cell in place of the LSTM, which we leave to future work.

III. METHOD

Multi-stream macro highway. We expand the residual stream into n parallel streams $X_l \in \mathbb{R}^{B \times T \times n \times d}$ (we use $n=4, d=128$). At each inter-layer boundary we flatten and RMS-normalize [18] the streams and compute three heads from the normalized input: an aggregation gate $\mathcal{H}^{pre} = \sigma(\cdot) \in [0, 1]$, a distribution gate $\mathcal{H}^{post} = 2\sigma(\cdot) \in [0, 2]$, and a residual-mixing matrix $\mathcal{H}^{res} = \text{SinkhornKnopp}(\cdot)$ projected onto the Birkhoff polytope. Each head combines a small input-dependent (dynamic) term, gated by a learnable α initialized to 0.01, with a learned static bias, following the mHC parameterization. The layer update is

$$X_{l+1} = \mathcal{H}_l^{res} X_l + \mathcal{H}_l^{post} \text{LSTM}_l(\sum_n \mathcal{H}_l^{pre} X_l). \quad (1)$$

The aggregation gate collapses the n streams into a single d -dimensional input for a standard LSTM layer; the distribution gate scatters the layer output back across streams; and the doubly-stochastic $\mathcal{H}^{res}$ mixes the streams as a convex (soft-permutation) combination.

Why macro placement (complexity). Because $\mathcal{H}^{res}$ is computed once per layer over the batched $B \times T$ tensor and the LSTM consumes the whole sequence in one cuDNN pass, the Sinkhorn–Knopp projection executes $O(L)$ times per forward pass. A *micro* placement that applies the projection inside the recurrent cell at every

timestep would instead execute it $O(T \cdot L)$ times. With $T=724$ this is a factor of T difference (L vs. $\sim 724L$ projections), and it also forgoes the cuDNN sequence kernel by requiring a step-by-step cell loop. We therefore adopt the macro placement; Sec. V-G formalizes this argument. We do not train a micro variant: the comparison here is analytical (operation count), not empirical.

A. Theory

The Birkhoff polytope $\mathcal{M}^{res} = \{H \geq 0 : H\mathbf{1}=\mathbf{1}, \mathbf{1}^\top H=\mathbf{1}^\top\}$ confers three properties that make depth stable. (i) *Non-expansiveness*: every doubly-stochastic H has spectral norm $\|H\|_2 \leq 1$, so the residual map cannot amplify signal energy. (ii) *Compositional closure*: the product of doubly-stochastic matrices is doubly-stochastic, so the composite map across any number of layers $\prod_l \mathcal{H}_l^{res}$ remains non-expansive — stability holds at arbitrary depth, not just per layer. (iii) *Convex hull of permutations*: the polytope is the convex hull of permutation matrices, so $\mathcal{H}^{res}$ acts as a soft permutation that fuses streams without destructive amplification. When $n=1$ the constraint reduces to the scalar 1, recovering the standard identity-mapping residual. Section V confirms these properties empirically: the measured composite spectral norm stays near 1.0 for mHC at every depth.

IV. EXPERIMENTAL SETUP

Datasets. We use three real multivariate series with chronological, gap-separated 70/15/15 train/val/test splits: the NOAA JFK hourly weather benchmark (75,119 rows, 4 variables—temperature, humidity, pressure, wind speed), ETTh1 (17,420 rows), and the larger ETTm1 (69,680 rows, 15-min cadence). This is the NOAA JFK weather series, not the 21-variable “Weather” benchmark used in many long-term-forecasting papers, so all baseline numbers reported here are from our own retraining under the protocol below. The z -score scaler is fit on the training split only; a guard gap equal to the maximum horizon separates splits so no window crosses a boundary.

Protocol. Every model uses an identical regimen: AdamW [19] ($\beta=(0.9, 0.95)$, $\epsilon=10^{-8}$, weight decay 0.1), learning rate 4×10^{-4} with 5-epoch linear warmup then cosine decay, dropout 0.1, gradient clipping at 1.0, batch size 32, up to 50 epochs with early stopping (patience 10), and five seeds (42–46) for the main weather/ETTh1 experiments. Forecast horizons are 6/12/24/48/72 hours. The only difference between models is the inter-layer highway. We select the checkpoint with the best validation MSE and evaluate it *once* on the held-out test split; all reported metrics are de-normalized (original units).

Models. Standard LSTM (parameter-matched), unconstrained HC-LSTM, and mHC-LSTM (ours), plus classical/linear baselines (Ridge, DLinear, NLinear) and neural baselines (GRU, TCN), at depths $L \in \{4, 8, 16\}$; depth $L=32$ adds the large-dataset comparison on weather and ETTm1.

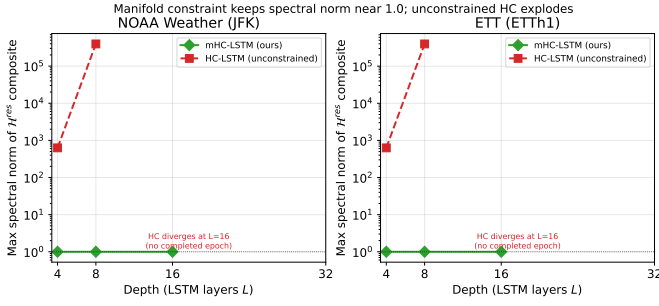


Fig. 1. Max composite spectral norm vs. depth. mHC stays near 1.0; HC explodes and then diverges at $L=16$.

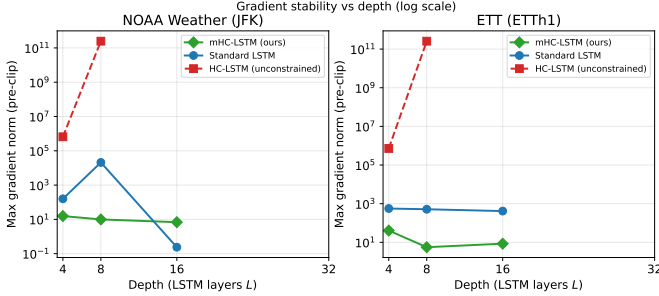


Fig. 2. Max pre-clip gradient norm vs. depth (log scale). mHC stays bounded (<50) at all depths; HC reaches 10^5 – 10^{11} and diverges at $L=16$.

V. RESULTS

A. Training stability

mHC keeps the composite residual spectral norm within numerical tolerance of 1.0, while unconstrained HC explodes (Table I, Fig. 1) and diverges (NaN) at $L=16$ on both datasets.

B. Depth scaling (the main result)

On held-out test data, mHC preserves stable error across the evaluated depth range; Standard LSTM and TCN collapse at $L \geq 16$ (Table II, Fig. 3). The stability comes at a modest absolute-cost increase: mHC’s test MSE at $L=16$ on weather (95.4) is higher than its own $L=4$ value (91.7), so deeper mHC does not improve raw accuracy on these tasks; rather, it avoids the catastrophic collapse seen in the vanilla LSTM.

Relation to stronger recurrent baselines. We note frankly that a plain GRU—a standard recurrent baseline with no highway machinery and fewer parameters (351K vs. 592K)—achieves lower weather test MSE than mHC-LSTM at every depth we evaluate (87.7/89.0/94.5 vs. 91.7/94.3/95.4 at $L=4/8/16$) and is statistically tied with mHC on ETTh1 (within 0.02–0.5 MSE). Ridge and the linear DLinear/NLinear baselines also outperform mHC on ETTh1. These results reframe the contribution: mHC-LSTM is not a generic state-of-the-art forecaster on these datasets at shallow depth. Its value is specific to the problem of training a deep stacked LSTM without collapse.

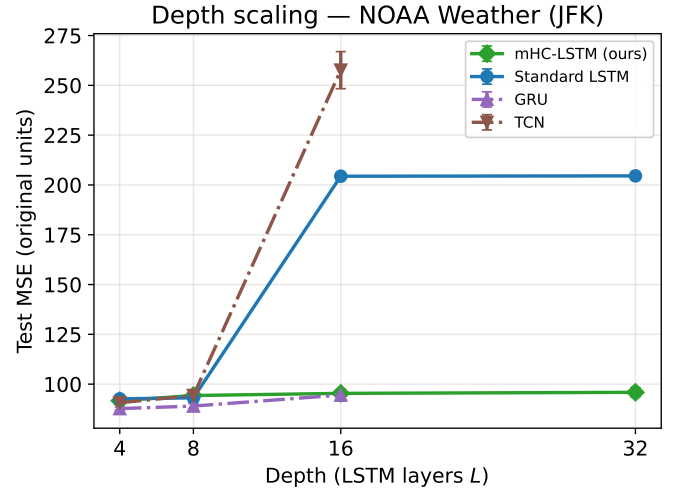


Fig. 3. Weather test MSE vs. depth. mHC stays in the 91.7–95.9 range across all depths; Standard LSTM and TCN collapse at $L \geq 16$.

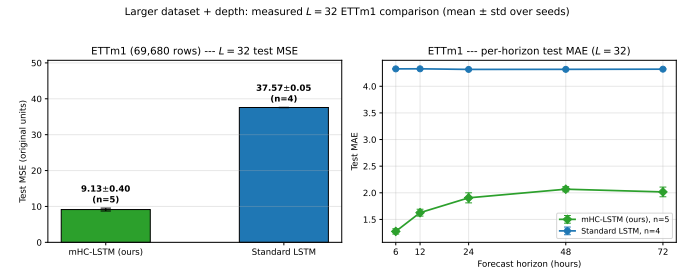


Fig. 4. ETTm1 $L=32$: mHC scales; Standard LSTM degrades.

Practically, this means mHC-LSTM is most interesting when an LSTM-based architecture is already preferred for deployment, interpretability, or hardware reasons, and one wishes to add depth without the instability that otherwise afflicts deep LSTMs.

C. Larger dataset (ETTm1)

On ETTm1 (69,680 rows) at $L=32$, mHC achieves 9.13 ± 0.40 test MSE (5 seeds) while Standard LSTM degrades to 37.57 ± 0.05 (4 seeds; one vanilla seed failed to complete and is excluded, not imputed), a $4.1\times$ gap (Fig. 4).

D. Baseline comparison and horizon degradation

At shallow depth ($L=4$) mHC is competitive with, but not statistically distinct from, the strongest classical, linear, recurrent, and convolutional baselines (Fig. 5); the benefit of mHC appears with depth rather than at $L=4$. Per-horizon test MAE degrades gracefully for mHC across the 6–72 h horizons, whereas collapsed deep baselines flatten to a near-constant error (Fig. 6).

E. Comparison with modern forecasters

We re-train three strong recent forecasters — PatchTST, iTransformer, and TimesNet — under our

TABLE I
TRAINING STABILITY (MAX OVER ALL STEPS/SEEDS). mHC HOLDS THE COMPOSITE SPECTRAL NORM NEAR 1.0 AT EVERY DEPTH;
UNCONSTRAINED HC EXPLODES (ALREADY AT $L=8$ ON ETTh1) AND DIVERGES (NaN) AT $L=16$.

Dataset	Model	L	Max grad norm	Max spec. norm
Weather	mHC-LSTM (ours)	4	1.56e+01	1.0000
Weather	HC-LSTM	4	6.56e+05	628.9461
Weather	Standard LSTM	4	1.59e+02	—
Weather	mHC-LSTM (ours)	8	9.92e+00	1.0000
Weather	HC-LSTM	8	2.47e+11	395483.5938
Weather	Standard LSTM	8	2.09e+04	—
Weather	mHC-LSTM (ours)	16	6.86e+00	1.0001
Weather	HC-LSTM	16	diverged	—
Weather	Standard LSTM	16	2.38e-01	—
ETTh1	mHC-LSTM (ours)	4	4.08e+01	1.0000
ETTh1	HC-LSTM	4	7.22e+05	629.1806
ETTh1	Standard LSTM	4	5.62e+02	—
ETTh1	mHC-LSTM (ours)	8	5.59e+00	1.0000
ETTh1	HC-LSTM	8	2.57e+11	397134.6875
ETTh1	Standard LSTM	8	5.14e+02	—
ETTh1	mHC-LSTM (ours)	16	8.50e+00	1.0001
ETTh1	HC-LSTM	16	diverged	—
ETTh1	Standard LSTM	16	4.14e+02	—

TABLE II
TEST MSE VS. DEPTH. mHC REMAINS STABLE ACROSS THE EVALUATED DEPTH RANGE; STANDARD LSTM AND TCN COLLAPSE AT DEPTH.

Dataset	Model	$L=4$	$L=8$	$L=16$	$L=32$
Weather	mHC-LSTM (ours)	91.7	94.3	95.4	95.9
Weather	Standard LSTM	92.6	93.1	204.4	204.6
Weather	GRU	87.7	89.0	94.5	—
Weather	TCN	90.7	94.3	257.6	—
ETTh1	mHC-LSTM (ours)	9.9	10.3	10.7	—
ETTh1	Standard LSTM	9.9	11.8	12.9	—
ETTh1	GRU	10.1	10.7	10.7	—
ETTh1	TCN	10.2	10.6	34.8	—

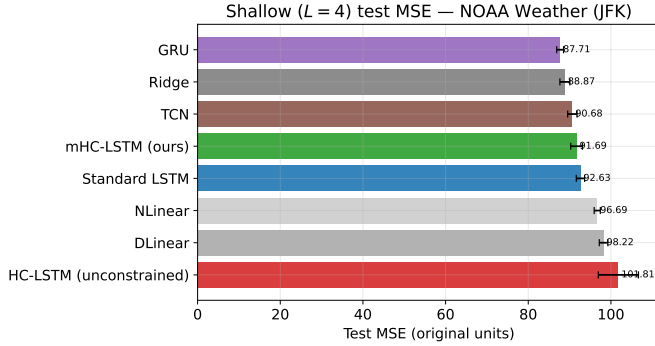


Fig. 5. Shallow ($L=4$) weather test MSE across all model families.

identical protocol (same splits, scaler, horizons, optimizer, 5 seeds, checkpoint selection), so the comparison is on our data, not copied numbers (Table V). On **weather**, mHC (91.7) significantly outperforms both PatchTST (95.9, $d=-3.5$, $p=0.001$) and iTransformer (98.5, $d=-2.3$, $p=0.007$) on test MSE (Table VI); TimesNet (96.2) is also significantly worse than mHC on this dataset ($d=-1.36$, $p=0.038$), though with higher seed variance ($\sigma=3.3$). Our

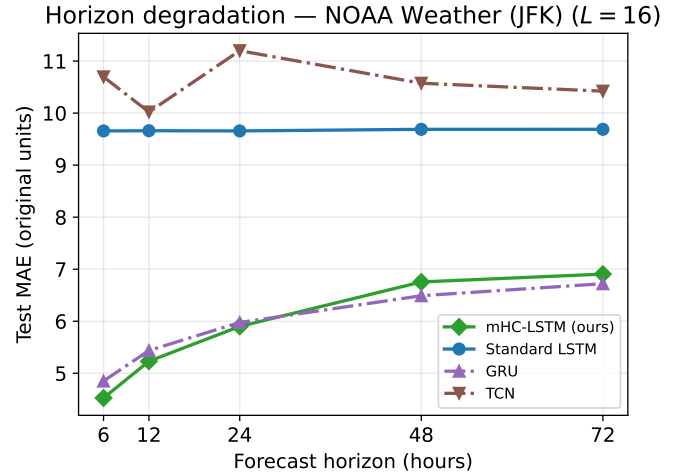


Fig. 6. Per-horizon test MAE at $L=16$ on weather. Standard LSTM flattens (near-constant prediction) while mHC degrades gracefully with horizon.

TimesNet baseline is reimplemented following the module structure of the Time-Series-Library benchmark [20]. On **ETTh1**, PatchTST (8.5) significantly beats mHC

TABLE III
HELD-OUT **test**-SET FORECASTING ERROR (ORIGINAL UNITS, MEAN $\pm$ STD OVER SEEDS). LOWER IS BETTER. HC DIVERGES (NAN) AT $L=16$;
VANILLA COLLAPSES AT $L\geq 16$.

Dataset	Model	L	Test MSE	Test MAE	n
Weather	mHC-LSTM (ours)	4	91.69 ± 1.41	5.729 ± 0.043	5
Weather	Standard LSTM	4	92.63 ± 1.01	5.823 ± 0.037	5
Weather	HC-LSTM	4	101.81 ± 4.86	6.012 ± 0.119	5
Weather	GRU	4	87.71 ± 0.83	5.609 ± 0.035	5
Weather	TCN	4	90.68 ± 1.13	5.722 ± 0.042	5
Weather	Ridge	4	88.87 ± 1.21	5.644 ± 0.053	5
Weather	DLinear	4	98.22 ± 1.03	6.056 ± 0.054	4
Weather	NLinear	4	96.69 ± 0.74	5.939 ± 0.013	5
Weather	mHC-LSTM (ours)	8	94.29 ± 0.85	5.820 ± 0.025	5
Weather	Standard LSTM	8	93.12 ± 2.30	5.883 ± 0.087	5
Weather	HC-LSTM	8	94.00 ± 2.71	5.795 ± 0.094	5
Weather	GRU	8	88.99 ± 1.19	5.672 ± 0.066	5
Weather	TCN	8	94.29 ± 0.98	5.909 ± 0.044	5
Weather	mHC-LSTM (ours)	16	95.37 ± 1.39	5.863 ± 0.040	5
Weather	Standard LSTM	16	204.38 ± 0.29	9.671 ± 0.005	5
Weather	GRU	16	94.53 ± 3.04	5.895 ± 0.108	5
Weather	TCN	16	257.63 ± 9.31	10.583 ± 0.156	4
Weather	mHC-LSTM (ours)	32	95.89 ± 2.44	5.878 ± 0.061	5
Weather	Standard LSTM	32	204.55 ± 0.39	9.672 ± 0.009	3
ETTh1	mHC-LSTM (ours)	4	9.92 ± 0.26	1.952 ± 0.046	5
ETTh1	Standard LSTM	4	9.92 ± 0.71	2.028 ± 0.089	5
ETTh1	HC-LSTM	4	44.52 ± 18.73	4.323 ± 0.983	5
ETTh1	GRU	4	10.08 ± 0.22	2.014 ± 0.046	5
ETTh1	TCN	4	10.16 ± 0.44	2.097 ± 0.109	5
ETTh1	Ridge	4	7.77 ± 0.09	1.691 ± 0.020	5
ETTh1	DLinear	4	8.35 ± 0.06	1.718 ± 0.011	5
ETTh1	NLinear	4	8.26 ± 0.08	1.712 ± 0.017	5
ETTh1	mHC-LSTM (ours)	8	10.26 ± 0.56	1.995 ± 0.129	5
ETTh1	Standard LSTM	8	11.79 ± 3.06	2.254 ± 0.351	5
ETTh1	HC-LSTM	8	$8605623.25 \pm 3713064.63$	1900.638 ± 334.541	5
ETTh1	GRU	8	10.74 ± 0.36	2.048 ± 0.049	5
ETTh1	TCN	8	10.57 ± 1.17	2.153 ± 0.110	5
ETTh1	mHC-LSTM (ours)	16	10.65 ± 0.66	2.058 ± 0.108	5
ETTh1	Standard LSTM	16	12.88 ± 2.60	2.378 ± 0.292	5
ETTh1	GRU	16	10.67 ± 0.48	2.116 ± 0.055	5
ETTh1	TCN	16	34.84 ± 4.88	4.168 ± 0.369	5
ETTM1	mHC-LSTM (ours)	32	9.13 ± 0.40	1.778 ± 0.037	5
ETTM1	Standard LSTM	32	37.57 ± 0.05	4.322 ± 0.005	4

(9.9), while mHC significantly beats iTransformer (10.7); TimesNet (9.4) is non-significantly better ($p=0.348$). Thus mHC is competitive with 2023–2024 SOTA — leading on weather, mid-pack on ETTh1 — which, combined with its unique depth-stability (Sec. VA–B), supports the contribution without overclaiming shallow accuracy.

F. Compute efficiency

mHC adds negligible parameter/FLOP overhead over a standard LSTM: at every depth the two are parameter- and FLOP-matched (e.g., at $L=4$, 592K vs. 590K parameters and 0.85 vs. 0.86 GFLOPs), since the $n=4$ stream heads are tiny relative to the LSTM matmuls (Table VII). The cost of the multi-stream highway is *activation memory*: mHC’s peak VRAM is higher than the standard LSTM because it carries n streams, and *CPU inference latency*: mHC (1539ms) is $\sim 3\times$ slower than the standard LSTM (508ms) due to sequential stream mixing, while iTransformer (3.4ms) benefits from parallel attention in this CPU benchmark. Parameters and FLOPs

are measured on a $1\times 724\times 4$ input; latency is mean over 100 CPU forward passes (batch=32). We do not claim GPU inference latency from this CPU measurement.

G. Macro vs. micro placement: a complexity argument

The doubly-stochastic projection can be placed either between layers (*macro*, ours) or inside the recurrent cell at every timestep (*micro*). We compare the two analytically. Let K be the Sinkhorn–Knopp iteration count and n the number of streams. Per forward pass over a length- T sequence with L layers, the macro placement performs L projections (each a batched $n\times n$ Sinkhorn over the $B\times T$ elements), i.e. $O(LKn^2)$ Sinkhorn work *outside* the temporal recurrence, so the LSTM still runs as a single fused cuDNN kernel. The micro placement performs one projection *per timestep per layer*, i.e. $O(TLKn^2)$, and forces a step-by-step LSTMCell loop that forgoes the cuDNN sequence kernel. For our setting ($T=724$) the micro placement issues $\sim 724\times$ more projections and loses hardware-optimized recurrence. We therefore adopt the

TABLE IV
 PAIRED SIGNIFICANCE ON **test** MSE FOR MATCHED FIVE-SEED COMPARISONS: COHEN’S d AND TWO-SIDED t -TEST p . NEGATIVE DIFF FAVORS mHC.

Dataset	Comparison	L	Δ MSE	Cohen’s d	p
ETTh1	mHC vs HC	4	-34.60	-1.84	0.015*
ETTh1	mHC vs Vanilla	4	-0.01	-0.01	0.989
ETTh1	mHC vs HC	8	-8605612.99	-2.32	0.007*
ETTh1	mHC vs Vanilla	8	-1.53	-0.51	0.314
ETTh1	mHC vs Vanilla	16	-2.23	-0.70	0.192
Weather	mHC vs HC	4	-10.12	-1.74	0.018*
Weather	mHC vs Vanilla	4	-0.94	-0.51	0.318
Weather	mHC vs HC	8	+0.29	0.08	0.860
Weather	mHC vs Vanilla	8	+1.17	0.41	0.406
Weather	mHC vs Vanilla	16	-109.01	-76.72	0.000*

* significant at $p < 0.05$.

TABLE V
 COMPARISON AGAINST MODERN FORECASTERS (HELD-OUT **test** MSE/MAE, MEAN $\pm$ STD OVER FIVE SEEDS). OUR LSTM MODELS ARE AT $L=4$; TRANSFORMER BASELINES USE THEIR OWN DEPTH. mHC IS BEST ON WEATHER; PATCHTST/NLINEAR LEAD ON ETTh1.

Model	ETTh1 MSE	ETTh1 MAE	Weather MSE	Weather MAE
NLinear	8.26 ± 0.08	1.712 ± 0.017	96.69 ± 0.74	5.939 ± 0.013
DLinear	8.35 ± 0.06	1.718 ± 0.011	98.22 ± 1.03	6.056 ± 0.054
Ridge	7.77 ± 0.09	1.691 ± 0.020	88.87 ± 1.21	5.644 ± 0.053
TCN	10.16 ± 0.44	2.097 ± 0.109	90.68 ± 1.13	5.722 ± 0.042
GRU	10.08 ± 0.22	2.014 ± 0.046	87.71 ± 0.83	5.609 ± 0.035
PatchTST	8.51 ± 0.34	1.833 ± 0.050	95.92 ± 1.05	5.971 ± 0.032
iTransformer	10.74 ± 0.52	2.065 ± 0.049	98.47 ± 1.72	6.116 ± 0.059
TimesNet	9.40 ± 0.86	1.998 ± 0.144	96.25 ± 3.34	5.932 ± 0.116
mHC-LSTM (ours)	9.92 ± 0.26	1.952 ± 0.046	91.69 ± 1.41	5.729 ± 0.043

TABLE VI
 PAIRED SIGNIFICANCE OF mHC ($L=4$) VS. SOTA ON **test** MSE (FIVE MATCHED SEEDS). NEGATIVE DIFF FAVORS mHC.

Dataset	vs.	Δ MSE	Cohen’s d	p
Weather	PatchTST	-4.23	-3.52	0.001*
Weather	iTransformer	-6.79	-2.30	0.007*
Weather	TimesNet	-4.56	-1.36	0.038*
ETTh1	PatchTST	+1.41	2.54	0.005*
ETTh1	iTransformer	-0.82	-1.76	0.017*
ETTh1	TimesNet	+0.52	0.48	0.348

* significant at $p < 0.05$.

macro placement on complexity grounds; we do not train a micro variant, so no empirical accuracy comparison is claimed.

VI. DISCUSSION AND LIMITATIONS

What the results support. The central claim — that doubly-stochastic inter-layer mixing yields stable, trainable depth for LSTMs — is supported on held-out test data: mHC’s error remains stable with depth, its composite spectral norm stays near 1.0, and its gradient norm stays bounded, whereas standard LSTM and TCN collapse at $L \geq 16$ and unconstrained HC diverges. The advantage over HC is statistically significant at $L=4$ on both datasets, and the advantage over standard LSTM at $L=16$ on weather is large ($d \approx -77$, $p < 10^{-3}$).

What we do not claim. At *shallow* depth mHC is statistically indistinguishable from a standard LSTM and from the strongest baselines on test data; this is a

stability/trainability result, not a claim of shallow state-of-the-art accuracy. **Statistical caveats.** All significance tests use five matched seeds. Large Cohen’s d values (e.g., $d \approx -77$ for weather $L=16$) reflect a large mean gap relative to the very small seed-to-seed variance in that paired comparison, but at $n=5$ such point estimates are fragile. We report uncorrected paired t -test p -values across roughly seventeen comparisons in Tables IV and VI; a few borderline claims ($p \approx 0.038$) would not survive a Holm–Bonferroni multiple-comparison correction. The robust conclusions are the qualitative stability pattern (spectral norm near 1.0, bounded gradients) and the large weather depth-collapse gap, not fine-grained ranking of methods within 1–2 MSE points.

Limitations. The ETTh1 result has 5 seeds for mHC and 4 for vanilla (one vanilla seed failed to complete and is excluded, not imputed). Our SOTA comparison covers

TABLE VII

COMPUTE EFFICIENCY (WEATHER). PARAMS AND FLOPS MEASURED ON A $1 \times 724 \times 4$ INPUT; PEAK VRAM AND TRAIN TIME FROM TRAINING LOGS; INFERENCE LATENCY ON CPU (BATCH=32, 100 RUNS, MEAN). MHC IS PARAMETER- AND FLOP-MATCHED TO THE STANDARD LSTM, AT HIGHER ACTIVATION MEMORY AND CPU LATENCY FROM THE n -STREAM EXPANSION.

Model	L	Params	GFLOPs	VRAM (MB)	Train/epoch (s)	Latency (ms)
mHC-LSTM (ours)	4	592,512	0.845	1539	182.8	1539.3
Standard LSTM	4	590,420	0.857	636	296.3	507.8
HC-LSTM	4	592,512	0.845	1249	149.3	1349.9
GRU	4	351,252	0.510	—	—	470.7
TCN	4	153,108	0.216	—	—	131.6
PatchTST	4	446,725	0.098	—	—	106.8
iTransformer	4	490,885	0.002	—	—	3.4
TimesNet	4	4,738,157	13.572	—	—	5457.9
DLinear	4	7,250	0.000	—	—	2.0
NLinear	4	3,625	0.000	—	—	0.3

PatchTST, iTransformer, and TimesNet (5 seeds each); extending to FEDformer and Autoformer, and to additional long-term-forecasting benchmarks, is the clearest next step. Two baseline cells within the main five-seed sweep have only four seeds each (weather DLinear $L=4$, weather TCN $L=16$; one seed failed to produce a log in each case), reported as $n=4$ in Table III rather than imputed. We built and unit-tested a micro (per-timestep) mHC-LSTM and staged a 15-run training ablation (3 depths $\times$ 5 seeds); these runs were not completed, so Sec. V-G’s comparison remains analytical only, with no empirical micro-model accuracy numbers reported anywhere in this paper. All models share the same learning-rate schedule across depths; we did not perform a per-depth learning-rate search for the Standard LSTM, so part of the depth-collapse effect could in principle be a tuning artifact rather than a fundamental depth limit. Re-tuning depth-specific LR is an open ablation.

VII. CONCLUSION

We presented a macro-level manifold-constrained Hyper-Connection highway for stacked LSTMs that keeps cross-stream mixing doubly-stochastic while preserving cuDNN-optimized recurrence. Across measured real-data experiments up to $L=32$, it trains stably and avoids the depth-collapse that afflicts standard LSTMs, TCNs, and unconstrained Hyper-Connections. In short, the manifold constraint removes the LSTM depth ceiling.

REPRODUCIBILITY

Metrics are de-normalized test-set values. The main weather/ETTh1 experiments use five seeds; smaller- n extensions are marked in the tables. Checkpoints are selected by best validation MSE and evaluated once on the held-out test split. Tables are generated directly from measured logs; missing/diverged runs are reported, never imputed. Code, configs, the training log manifest, and the ‘make_latex_tables.py’ script will be released publicly upon acceptance; until then they are available from the authors on request.

REFERENCES

- [1] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [2] Y. Bengio, P. Simard, and P. Frasconi, “Learning long-term dependencies with gradient descent is difficult,” *IEEE Transactions on Neural Networks*, vol. 5, no. 2, pp. 157–166, 1994.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *CVPR*, 2016.
- [4] —, “Identity mappings in deep residual networks,” in *ECCV*, 2016.
- [5] D. Zhu, H. Huang, Z. Huang, Y. Zeng, Y. Mao, B. Wu, Q. Min, and X. Zhou, “Hyper-connections,” in *International Conference on Learning Representations (ICLR)*, 2025, arXiv:2409.19606.
- [6] Z. Xie, Y. Wei, H. Cao, C. Zhao, C. Deng, J. Li, D. Dai, H. Gao, J. Chang, K. Yu, L. Zhao, S. Zhou, Z. Xu, Z. Zhang, W. Zeng, S. Hu, Y. Wang, J. Yuan, L. Wang, and W. Liang, “mhc: Manifold-constrained hyper-connections,” *arXiv preprint arXiv:2512.24880*, 2025.
- [7] R. Sinkhorn and P. Knopp, “Concerning nonnegative matrices and doubly stochastic matrices,” *Pacific Journal of Mathematics*, vol. 21, no. 2, pp. 343–348, 1967.
- [8] DeepSeek-AI, “Deepseek-v3 technical report,” *arXiv preprint arXiv:2412.19437*, 2024.
- [9] A. Mutlu, Ş. Doğan, and T. Tuncer, “mHC-SSM: Manifold-constrained hyper-connections for state space language models with stream-specialized adapters,” *arXiv preprint arXiv:2605.08300*, 2026.
- [10] A. Zeng, M. Chen, L. Zhang, and Q. Xu, “Are transformers effective for time series forecasting?” in *AAAI*, 2023.
- [11] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, “Learning phrase representations using rnn encoder-decoder for statistical machine translation,” in *EMNLP*, 2014.
- [12] S. Bai, J. Z. Kolter, and V. Koltun, “An empirical evaluation of generic convolutional and recurrent networks for sequence modeling,” *arXiv preprint arXiv:1803.01271*, 2018.
- [13] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, “Informer: Beyond efficient transformer for long sequence time-series forecasting,” in *AAAI*, 2021.
- [14] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, “A time series is worth 64 words: Long-term forecasting with transformers,” in *ICLR*, 2023.
- [15] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, and M. Long, “itransformer: Inverted transformers are effective for time series forecasting,” in *ICLR*, 2024.
- [16] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, “Timesnet: Temporal 2d-variation modeling for general time series analysis,” in *ICLR*, 2023.
- [17] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *Conference on Language Modeling (COLM)*, 2024, arXiv:2312.00752.
- [18] B. Zhang and R. Sennrich, “Root mean square layer normalization,” in *NeurIPS*, 2019.

- [19] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *ICLR*, 2019.
- [20] Y. Wang, H. Wu, J. Dong, Y. Liu, C. Wang, M. Long, and J. Wang, “Deep time series models: A comprehensive survey and benchmark,” *arXiv preprint arXiv:2407.13278*, 2024.